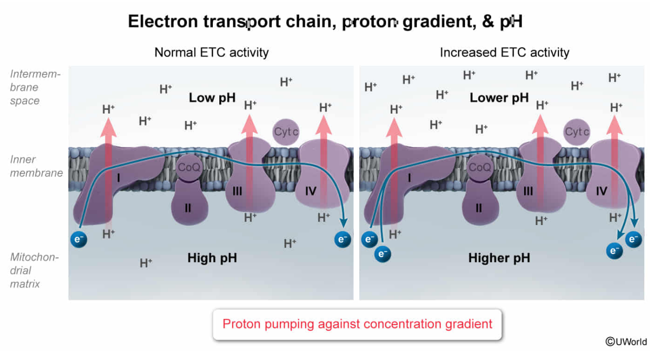
Which of the following events could directly increase the pH of the mitochondrial matrix?

- A. Permeabilization of the inner mitochondrial membrane [25%]
- B. Increased electron transport chain activity [38%]
- C. Increased ATP synthase activity [17%]
- D. Inhibition of the citric acid cycle [18%]

Correct
38% answered correctly

Explanation:



ATP synthesis in the mitochondria depends on a proton gradient across the inner mitochondrial membrane. This gradient is generated by the electron transport chain, which pumps protons out of the mitochondrial matrix and into the intermembrane space.

The proton gradient gives rise to the **proton motive force**, a chemiosmotic potential consisting of two components: a charge difference and a concentration difference across the inner mitochondrial membrane. The proton motive force drives protons back into the mitochondrial matrix through ATP synthase and provides the energy needed for ATP synthesis.

The pH of an environment is directly related to the proton concentration. A **lower proton concentration** corresponds to a **higher pH** and vice versa. Therefore, as **protons are pumped out** of the mitochondrial matrix, the **pH of the matrix increases**. Consequently, the **increased proton pumping** associated with **increased electron transport chain activity** would result in a **higher mitochondrial matrix pH**.

(Choice A) The inner mitochondrial membrane is generally impermeable to protons, and protons enter the matrix primarily through ATP synthase.

Membrane permeabilization and uncoupling proteins allow protons to flow down their gradient into the mitochondrial matrix at a higher rate, which *decreases* the matrix pH.

(Choice C) Protons flow *into* the mitochondrial matrix through ATP synthase. Increased ATP synthase activity would result in more protons flowing into the matrix, which would *decrease* the pH.

(Choice D) The citric acid cycle produces a substantial number of the NADH and $FADH_2$ molecules used in the electron transport chain. Inhibiting the citric acid cycle would result in fewer NADH and $FADH_2$ molecules entering the chain, which would in turn result in fewer protons being pumped out of the matrix. Ultimately, this would decrease the matrix pH.

Educational objective:
The electron transport chain pumps protons out of the mitochondrial matrix, resulting in a pH increase within the matrix. The proton motive force drives protons back into the matrix through ATP synthase, providing the energy for ATP synthesis.

Time spent:0 sec
Subject: Biochemistry

QID:403379
System: Metabolic Reactions